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Distributed Orchestration and Architecture Evolution for Edge-Cloud Collaborative AI Inference

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Bottlenecks of Pure Cloud AI Inference

Unbearable Latency

Round-trip time (RTT) to the cloud is typically above 100ms, which cannot meet the millisecond-level response required by autonomous driving and industrial robotics.

Bandwidth Scarcity

Uploading massive HD video streams and sensor data to the cloud in real time causes network congestion, and the transmission cost often exceeds the compute cost.

Data Privacy Risk

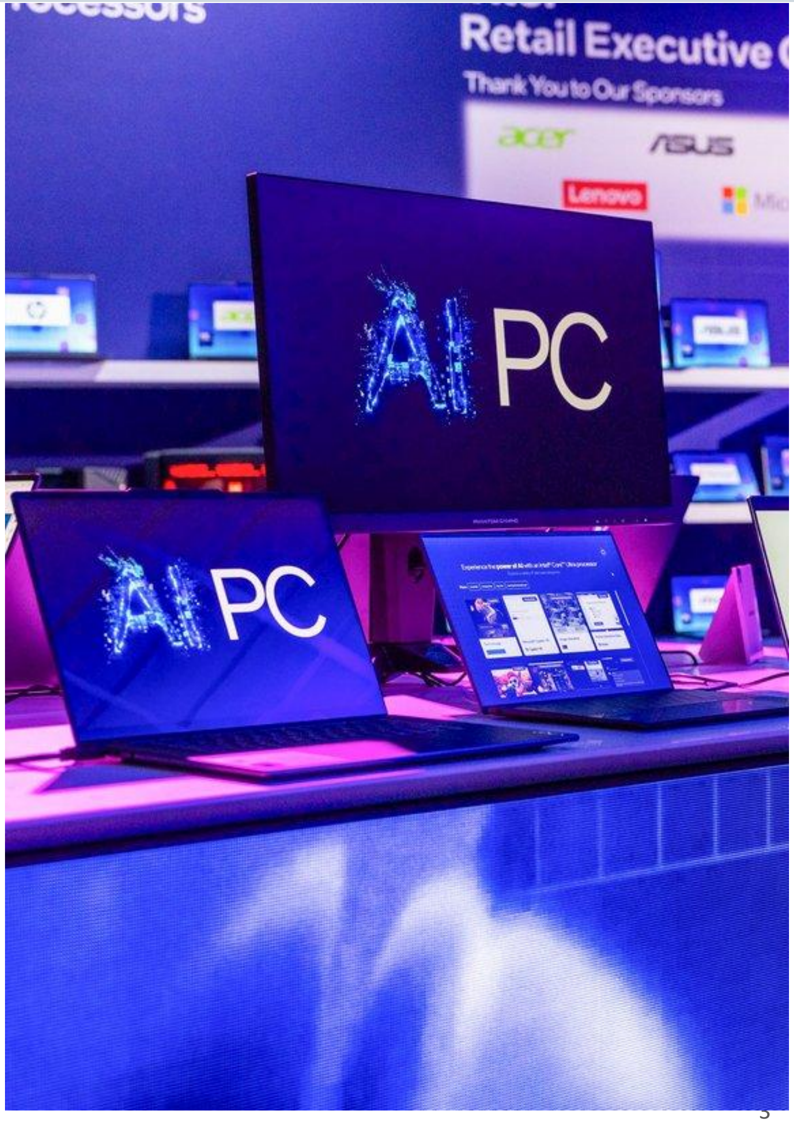
Sensitive data (personal privacy, trade secrets) faces leakage risk during public-network transmission and cloud storage, violating compliance requirements.

High Dependence on the Network

Once the network is disconnected, local terminals lose their intelligent processing capability, leading to service outage or safety incidents.

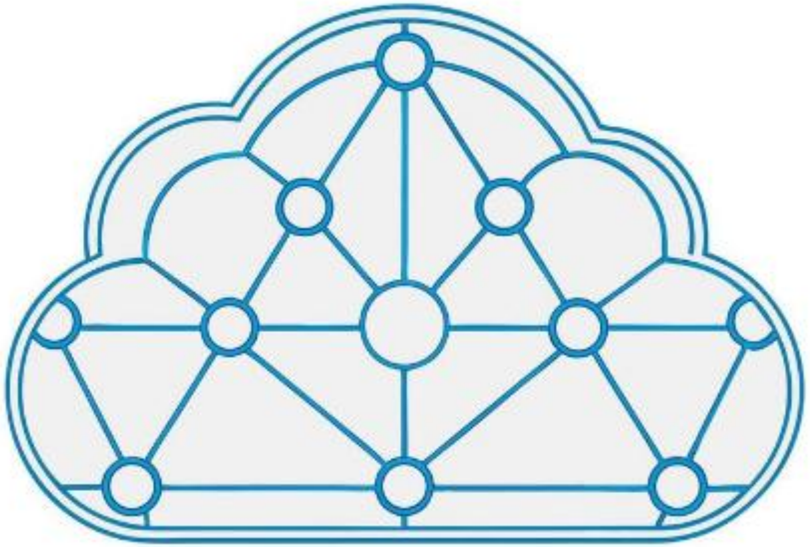
Limitations of Pure Edge/Device Inference

- Limited compute: Mobile devices and embedded chips can hardly run large deep-learning models (e.g., GPT-4, Llama-3).
- Accuracy loss: To fit device compute, models are heavily pruned and quantized, degrading inference accuracy.
- Maintenance difficulty: Edge nodes scattered across locations are hard to update, monitor, and troubleshoot in a unified way.
- Energy constraint: Terminal devices are often battery-limited; always-on high-power inference severely shortens battery life.



Cloud-Edge-Device Collaboration: The Wisdom of Divide and Conquer

Cloud: Responsible for large-scale model training, long-term historical data analysis, global scheduling, and complex logical reasoning.



Device: Real-time sensor data acquisition, ultra-low-latency simple inference, and serving as the interaction window.

Edge: Deployed in nearby server rooms, handling regional real-time prediction, filtering invalid data, and reducing cloud load.

The Inflection: From LLMs to Multi-Agent Systems

- LLMs are evolving from single-shot responders into Multi-Agent Systems (MAS): always-on, stateful, tool-calling, multi-turn, increasingly multimodal.
- Agentic workloads demand sustained, interactive, latency-bound inference — not the batch queries the cloud was originally sized for.
- Three forces collide at once:

– Latency control — closed-loop interaction tolerates no long round-trips.

– Privacy & data sovereignty — sensitive context must not always leave the device.

– Frontier-model reasoning — still too large for any single edge device.

- Implication: real-time control, privacy and heavy reasoning can no longer co-locate in one tier — the topology itself must change.

WHY IT MATTERS

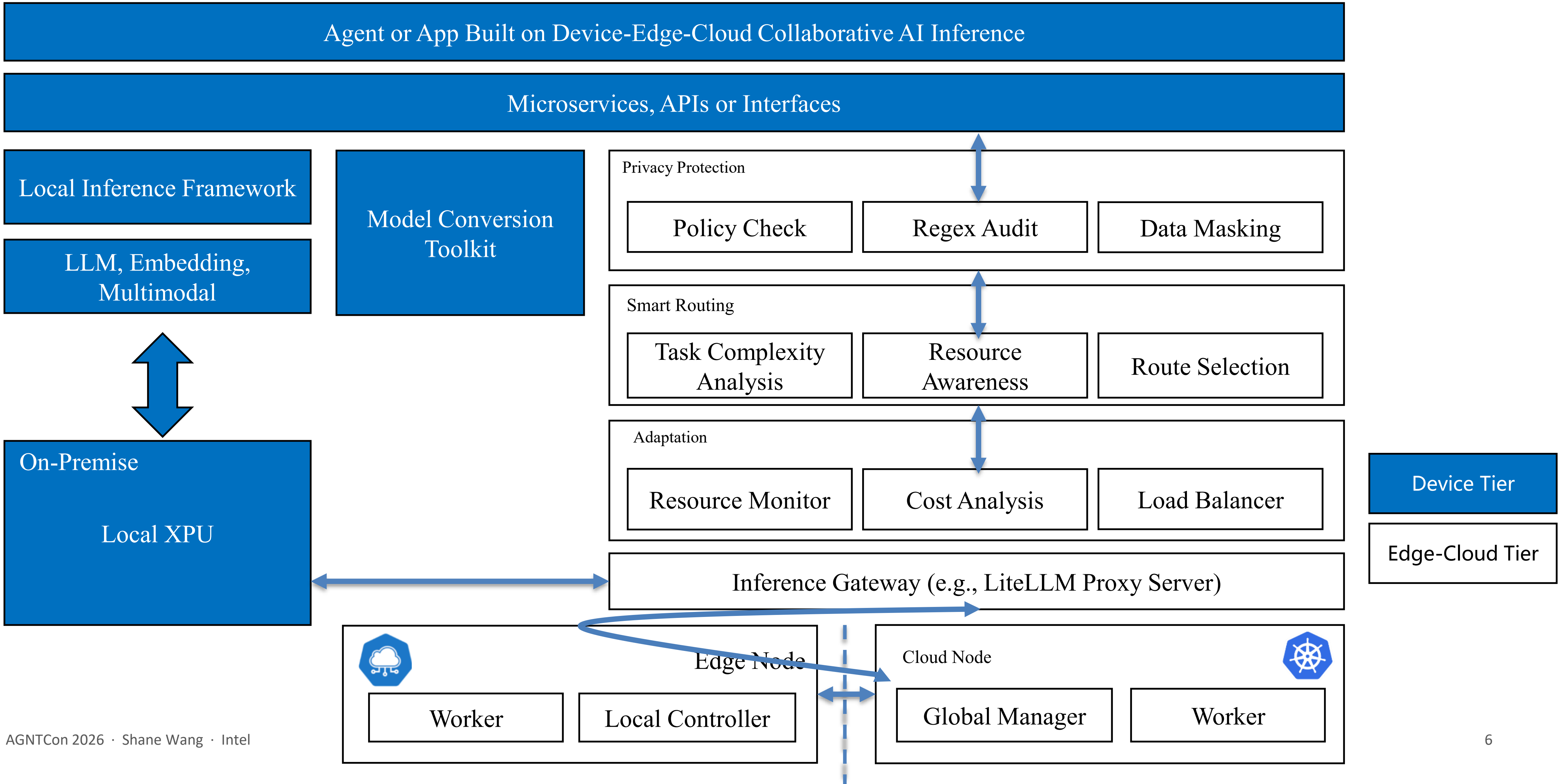
By 2028, ~33% of enterprise software applications will embed agentic AI, up from <1% today.

— Gartner (https://www.gartner.com/en/newsroom/press-releases/2025-06-25-gartner-predicts-over-40-percent-of-agentic-ai-projects-will-be-canceled-by-end-of-2027)

Agentic workloads are not just bigger queries — they are continuously-running, interactive inference with hard latency and privacy SLOs.

Reference Architecture for Device-Edge-Cloud Collaborative Inference

https://oaaif.org/agentos-omnicompute.html



Engineering Pillars

01
Heterogeneous Compute & Graph Partitioning

- Mixed-Silicon Synergy: Match Xeon, Gaudi, and NPU to workloads for optimal cost / perf / watt.
- Runtime Graph Partitioning: Split model graph & weights dynamically across Cloud, Edge, and Device.
- Frontier-Scale Execution: Run large models exceeding single-device memory while keeping low latency.

02
Network-Aware Scheduling

- Real-Time Telemetry: Continuously monitors latency, jitter, and node load to feed the scheduling engine.
- Dynamic Execution Shift: Microsecond-level shifting between edge and cloud as conditions change.
- SLO & Fail-Safe Routing: Routes by latency SLOs and auto-degrades gracefully if cloud link stalls.

03
Disconnected Self-Healing

- Pre-Staged Fallback: Keeps warm offline inference packs at edge for zero-downtime network drops.
- OTA Delta Patching: Ships lightweight incremental weight deltas from cloud instead of full models.
- Continuous Continuity: Keeps mission-critical operations (factories, vehicles) running and evolving.

Functional Partitioning: A Request's Journey

USER
Intent

→

DEVICE
NPU intent filtration
wake / biometric auth

→

EDGE
local RAG + PII sanit.
regional context

→

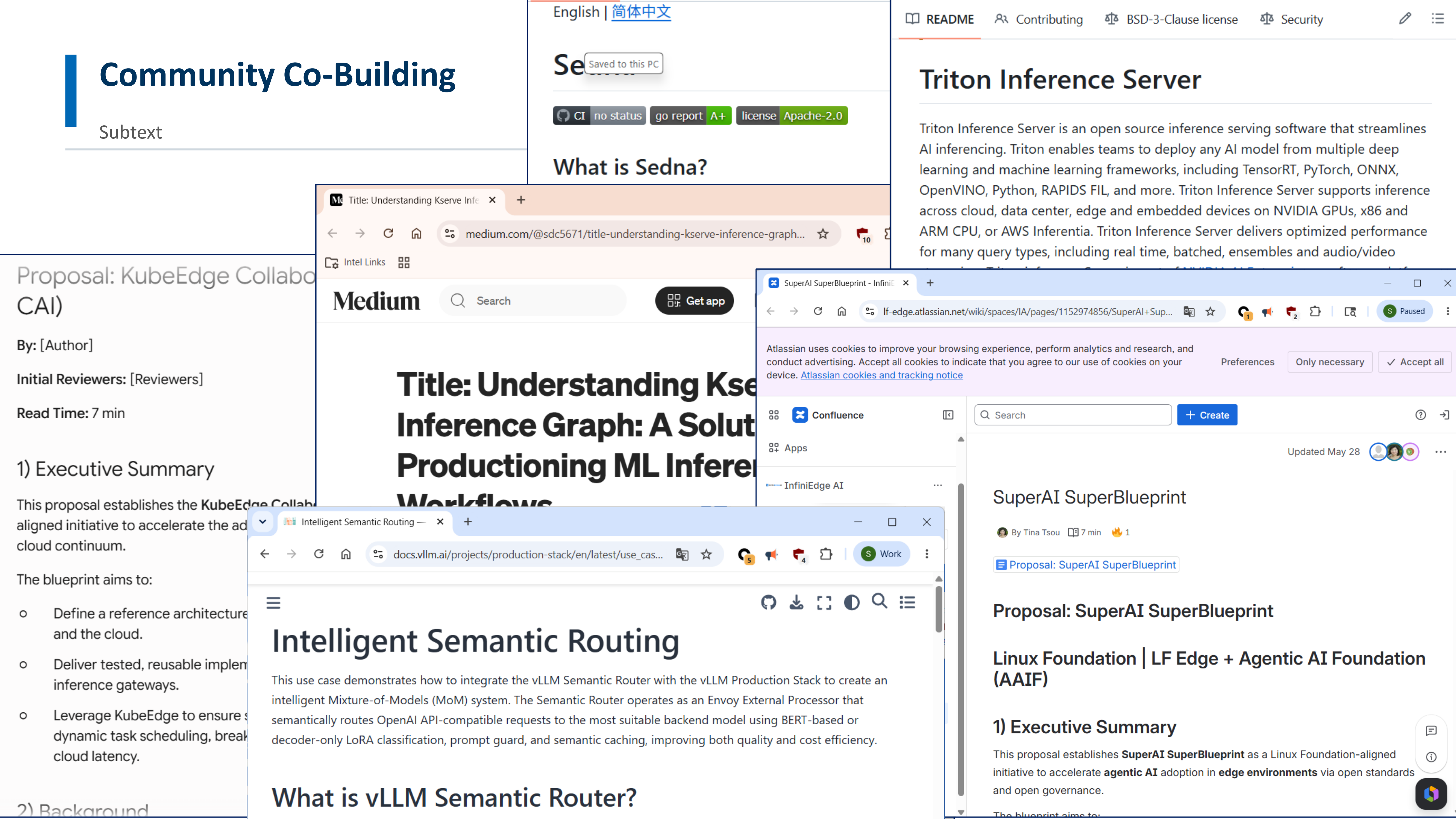
CLOUD
heavy reasoning
long-horizon plan

Response routed back the same path; raw personal data never leaves the device — only anonymized features / final results traverse to the cloud.

Concrete example — Personal AI assistant: on-device NPU filters intent and authenticates; edge keeps the local vector store and strips PII; cloud is only invoked for heavy multi-step reasoning and cross-domain knowledge.

Community Co-Building

Subtext



Advantages: Millisecond-Level Real-Time Response

<20ms

Typical end-to-end latency of collaborative inference

- By sinking inference tasks to the edge node closest to the data source, latency from long-haul fiber transmission is eliminated. This is a life- and property-critical metric in high-speed autonomous driving and precision industrial production.

  - Shorter path: processed locally, bypassing the core network
  - Local feedback: millisecond-level closed-loop control
  - Predictive execution: locally preloading likely-needed model fragments

Advantages: Strengthening Privacy and Data Sovereignty

Collaborative inference allows sensitive data to be processed locally, uploading only desensitized feature vectors or inference results to the cloud for secondary optimization. This mechanism perfectly fits legal frameworks such as the Personal Information Protection Law (PIPL) and the General Data Protection Regulation (GDPR).

> *"Over 50% of enterprises adopt edge computing to address data security and compliance issues." — Industry Survey Report*

Advantages: Extreme Resource Optimization and Cost Reduction

Collaborative inference can cut cloud compute bills by about 35%-40% and reduce backbone traffic by over 80%, greatly improving return on investment.

Advantages: Offline Autonomy and High Reliability

Uninterrupted Service

When the cloud link is down, the edge sustains basic intelligent services via a pre-installed "offline inference package," ensuring uninterrupted production.

Intelligent Dynamic Migration

Based on current network quality and node pressure, the system dynamically shifts the inference center of gravity between cloud and edge.

Incremental Model Update

The cloud retrains on aggregated hard samples and delivers only lightweight incremental weight patches, achieving continuous evolution.

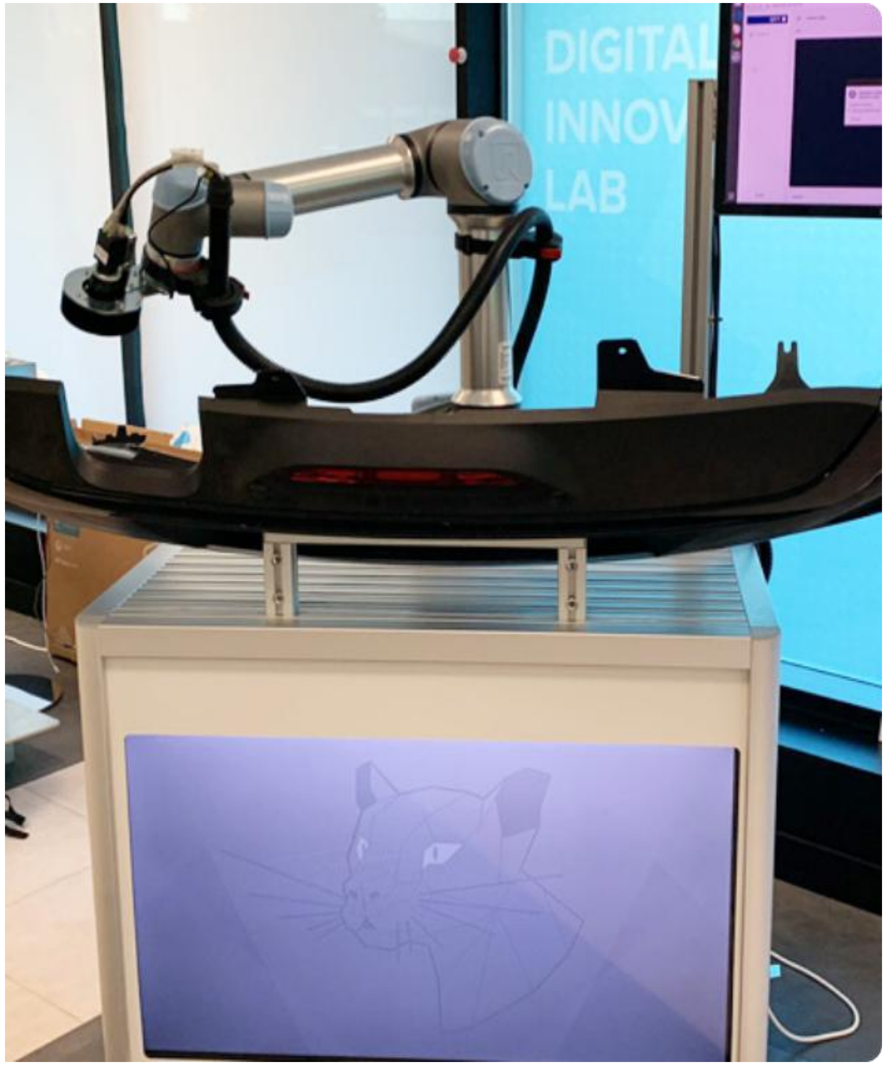
Scenarios - Personal AI: Intelligent Assistant

- **Paradigm shift**: evolving toward localized, proactive, and strongly context-aware personal AI agents.
- **Core pain point**: Traditional pure-cloud solutions face severe privacy/compliance risks, poor experience from high network latency, and consume massive bandwidth for environment-state synchronization.
- **Collaborative division of labor**:
  - Device: Runs lightweight models for instant wake-up, biometric authentication, and raw intent filtering.
  - Edge: Manages local home/personal vector stores, performs PII desensitization and local security interception.
  - Cloud: Intervenes only in heavy logical reasoning, long-range planning, and cross-domain knowledge integration.
- **Core value**: Building a zero-trust data-privacy barrier and enabling ultra-fast, millisecond-level, seamless intelligent companionship.

Scenarios - Robotics and Embodied Intelligence

- **Paradigm shift**: Robots are evolving toward embodied intelligence, requiring real-time closed loops of perception, reasoning, and action in highly dynamic physical spaces.
- **Core pain point**: Onboard compute limits multimodal model scale; closed-loop physical control cannot tolerate public-network latency; heavy reliance on wireless networks easily causes downtime.
- **Collaborative division of labor**:
  - Device: The low-level system performs high-frequency motion control, real-time SLM, and reflex-level obstacle avoidance.
  - Edge: Edge servers run large vision-planning models to coordinate overall planning.
  - Cloud: Acts as the central brain, responsible for complex simulation training, foundation-model iteration, and efficiency analysis.
- **Core value**: Breaking the single-machine compute ceiling and guaranteeing absolute mission autonomy under extreme network-disconnection conditions.

Scenarios - Smart Manufacturing: Defect Detection



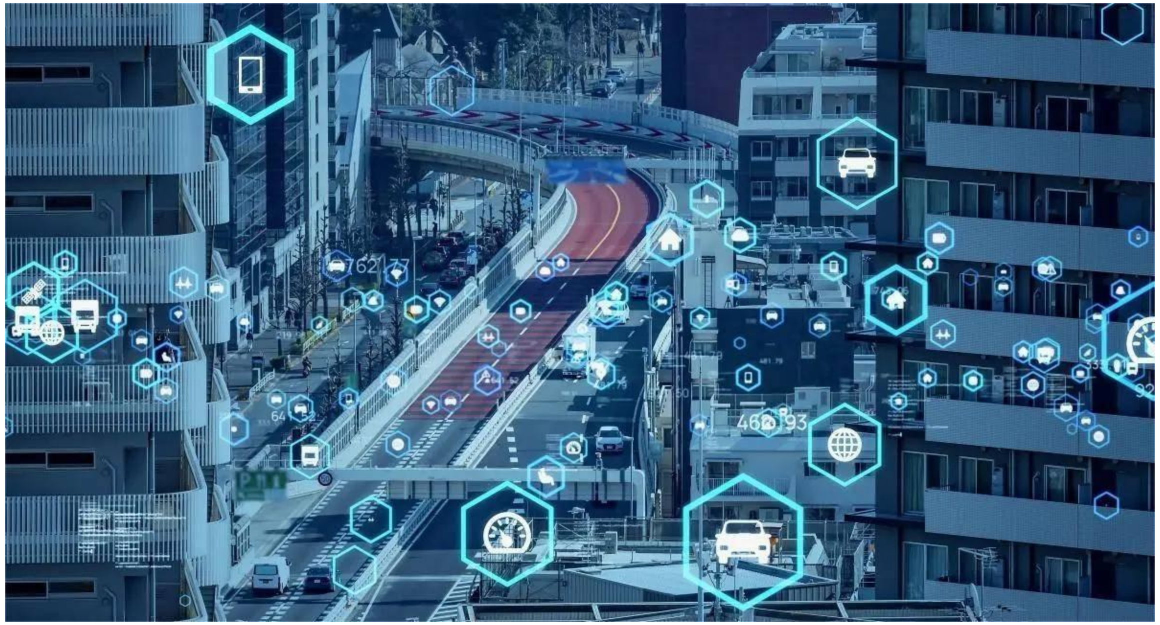
The Millisecond Contest of Industrial Vision

- On a high-speed conveyor, cameras capture hundreds of images per second. The device performs initial screening and simple cropping; the edge runs mid-size vision models to recognize scratches and cracks in real time and drive robotic arms to remove them. The cloud aggregates all defect images, optimizes detection accuracy, and performs production-dashboard analysis.
    - Response speed: <5ms
    - Bandwidth saving: 95% (only anomalous frames uploaded)
    - Effect: Miss rate reduced by 20%

Scenarios - Smart City: Traffic Scheduling

The Smart Brain of Citywide Perception

- Roadside perception devices (edge nodes) process intersection monitoring in real time, recognize vehicle types and flow, and dynamically adjust traffic-light timing. The cloud handles global topology planning and dispatches cross-intersection green-wave strategies to each edge node.
    - Real-time optimization: intersection congestion reduced by 25%
    - Data backhaul: supports post-hoc tracing and digital twins
    - Edge autonomy: network disconnection does not affect basic traffic-light logic



Scenarios - Autonomous Driving: Collaborative Perception

Single-Vehicle Intelligence + V2X Roadside Collaboration

- Vehicles (device) handle emergency obstacle avoidance via radar and cameras. Roadside units (edge) provide over-the-horizon vision, alerting the vehicle to pedestrians hidden by occlusion. The cloud maintains HD-map updates and long-range route planning.
    - Vision extension: eliminating single-vehicle blind spots
    - Reliable communication: ultra-low latency based on 5G-V2X
    - Intelligent scheduling: platooning control

Example on AI Hub

Applications on AI Hub – See Our Solutions at Intel Demo Booth

AI 方案展示中心

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Thank you.